Problem-Solution Fit Canvas

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence (CONSTRUCTAI)	**Maximum Marks:**	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]

• **Primary:** On-site Civil Safety Officers (EHS) & Lead QC Engineers.
• **Secondary:** Construction Project Managers & Planning Leads.
• **Tertiary:** Structural Consulting Engineers & Inspection Auditors.

6. CUSTOMER LIMITATIONS [CL]

• **Budget:** High-end IoT sensor grids are cost-prohibitive for standard job sites.
• **Devices:** Basic outdoor Android tablets, smartphones, and low-res CCTV feeds.
• **Connectivity:** Intermittent remote site cellular signals; zero on-site GPU servers.

5. AVAILABLE SOLUTIONS [AS]

• **Manual Visual Audits:**
Pros: Zero software expense.
Cons: High human fatigue, blind spots, zero real-time alerts.
• **Standard CCTV:**
Pros: 24/7 video capture.
Cons: Passive recording only, no automated hazard tagging or IS 456 checks.

2. PROBLEMS / PAINS + FREQ [PR]

• **PPE Violations:** Workers ditch helmets/vests 20+ times daily in humid conditions.
• **Crack Detection Lag:** Weekly delays diagnosing structural load-bearing cracks.
• **Weather Halts:** Sudden unpredicted rain ruins fresh concrete pours (bi-weekly in monsoon).

9. PROBLEM ROOT / CAUSE [RC]

• Inability of limited safety staff to monitor 360-degree multi-acre active sites.
• Lack of on-site structural expertise to instantly classify shear vs shrinkage cracks.
• Disconnected data: Weather radar, safety logs, and BOQ sheets are completely siloed.

7. BEHAVIOR + INTENSITY [BE]

• **High Intensity:** Panic-checking site when client or government auditors arrive.
• **Daily Routine:** Walking 15,000+ steps across dusty slabs taking phone photos.
• **Late Nights:** Manual paper log-keeping and WhatsApp status typing.

3. TRIGGERS TO ACT [TR]

• **Near-Miss Incidents:** A tool falls from height narrowly missing an un-helmeted worker.
• **Audit Notice:** Surprise municipal inspection risking site work-stop order.
• **Visible Wall Cracks:** Urgent client escalations regarding foundation settlement.

10. YOUR SOLUTION (CONSTRUCTAI) [SL]

Multimodal AI Construction Intelligence:
• **Groq LLaMA 3.2 Vision:** Sub-second detection of missing helmets, vests, and safety harnesses from images.
• **IS 456 Structural Crack Diagnosis:** Instant classification and chemical repair blueprints.
• **Automated Weather & BOQ Planner:** Predictive rain advisory and takeoff generator.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE:
• Web dashboard on mobile/tablet browsers.
• Automated PDF reports sent via email.
• Groq Vision Cloud API calls.
OFFLINE:
• Morning toolbox briefings & physical audit sheets.
• Local audible warning sirens on high-risk zones.

4. EMOTIONS BEFORE / AFTER [EM]

• **BEFORE:** Anxious, paranoid about fatal site accidents, overwhelmed by paper logs, and terrified of regulatory fines.
• **AFTER:** Confident, proactive, relieved with automated compliance logs, verified structural safety, and zero penalties.